

19-206-0711 ENERGY MANAGEMENT AND CONSERVATION

Module – 1

Basics of energy, its various forms
and conservation

Introduction



- Energy
 - The ability to do work
- Work
 - The transfer of energy from one form to another.
- In practical terms, energy is what we use to manipulate the world around us, whether by exciting our muscles, by using electricity, or by using mechanical devices such as automobiles.
- Energy comes in different forms - heat (thermal), light (radiant), mechanical, electrical, chemical, and nuclear energy.

Various Forms of Energy



- Two types of energy
 - Stored (potential) energy
 - Working (kinetic) energy
- Potential Energy – Stored energy and the energy of position.
 - Chemical Energy
 - Chemical energy is the energy stored in the bonds of atoms and molecules.
 - Biomass, petroleum, natural gas, propane and coal are examples of stored chemical energy.
 - Nuclear Energy
 - Nuclear energy is the energy stored in the nucleus of an atom - the energy that holds the nucleus together.
 - The nucleus of a uranium atom is an example of nuclear energy.

Various Forms of Energy



- **Stored Mechanical Energy**
 - Stored mechanical energy is energy stored in objects by the application of a force.
 - Compressed springs and stretched rubber bands are examples of stored mechanical energy.
- **Gravitational Energy**
 - Gravitational energy is the energy of place or position.
 - Water in a reservoir behind a hydropower dam is an example of gravitational energy. When the water is released to spin the turbines, it becomes motion energy.
- **Kinetic energy – Energy in motion**
 - **Radiant Energy**
 - Radiant energy is electromagnetic energy that travels in transverse waves.

Various Forms of Energy

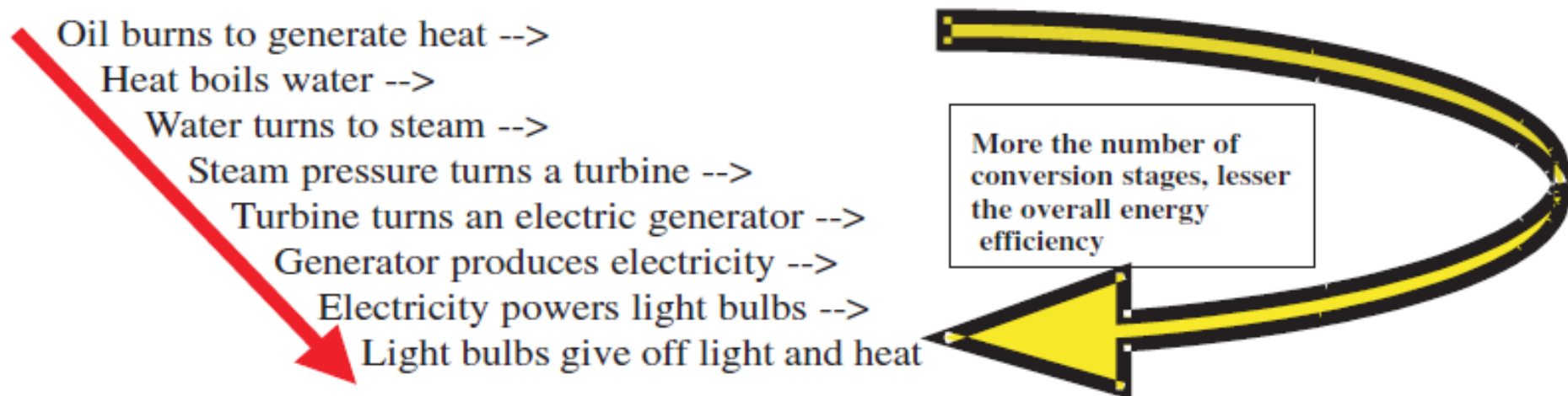


- Radiant energy includes visible light, x-rays, gamma rays and radio waves.
- Solar energy is an example of radiant energy.
- Thermal Energy
 - Thermal energy (or heat) is the internal energy in substances- the vibration and movement of atoms and molecules within substances.
 - Geothermal energy is an example of thermal energy.
- Motion
 - The movement of objects or substances from one place to another is motion.
 - Wind and hydropower are examples of motion.
- Sound
 - Sound is the movement of energy through substances in longitudinal (compression/rarefaction) waves.
- Electrical Energy
 - Electrical energy is the movement of electrons.
 - Lightning and electricity are examples of electrical energy.-

Various Forms of Energy



- Energy Conversion
 - Energy – ability to do work
 - Examples of work
 - Moving something
 - Lifting something
 - Warming something
 - Lighting something



Various Forms of Energy



- Grades of Energy

- High-Grade Energy

- Electrical and chemical energy are high-grade energy, because the energy is concentrated in a small space.
 - Even a small amount of electrical and chemical energy can do a great amount of work.
 - The molecules or particles that store these forms of energy are highly ordered and compact and thus considered as high grade energy.
 - High-grade energy like electricity is better used for high grade applications like melting of metals rather than simply heating of water.

- Low-Grade Energy

- Heat is low-grade energy.
 - Heat can still be used to do work (example of a heater boiling water), but it rapidly dissipates.
 - The molecules, in which this kind of energy is stored (air and water molecules), are more randomly distributed than the molecules of carbon in a coal.
 - This disordered state of the molecules and the dissipated energy are classified as low-grade energy.

Electrical Energy Basics



- Electric current is divided into two types:
 - Directional Current (DC)
 - Alternating Current (AC)
- Directional (Direct) Current
 - A non-varying, unidirectional electric current (Example: Current produced by batteries)
 - Characteristics:
 - Direction of the flow of positive and negative charges does not change with time
 - Direction of current (direction of flow for positive charges) is constant with time
 - Potential difference (voltage) between two points of the circuit does not change polarity with time
- Alternating Current
 - A current which reverses in regularly recurring intervals of time and which has alternately positive and negative values, and occurring a specified number of times per second.
 - Example: Household electricity produced by generators, Electricity supplied by utilities.

Electrical Energy Basics



- Characteristics:
 - Direction of the current reverses periodically with time .
 - Voltage (tension) between two points of the circuit changes polarity with time. .
 - In 50 cycle AC, current reverses direction 100 times a second (two times during one cycle)
- Ampere (A)
 - Current is the rate of flow of charge.
 - The ampere is the basic unit of electric current.
 - It is that current which produces a specified force between two parallel wires, which are 1 metre apart in a vacuum.
- Voltage (V)
 - The volt is the International System of Units (SI) measure of electric potential or electromotive force.
 - A potential of one volt appears across a resistance of one ohm when a current of one ampere flows through that resistance.

Electrical Energy Basics



- Resistance
 - Resistance = Voltage/Current
 - The unit of resistance is ohm (Ω)
- Ohm' Law
 - Ohm's law states that the current through a conductor is directly proportional to the potential difference across it, provided the temperature and other external conditions remain constant.
- Frequency
 - The supply frequency tells us the cycles at which alternating current changes.
 - The unit of frequency is hertz (Hz – cycles per second).
- Apparent power, kilovolt ampere (kVA)
 - It is the product of kilovolts and amperes.
 - This measures the **electrical load** on a circuit or system.
 - For a single phase electrical circuit, Apparent power (kVA) = Voltage x Amperes / 1000

Electrical Energy Basics



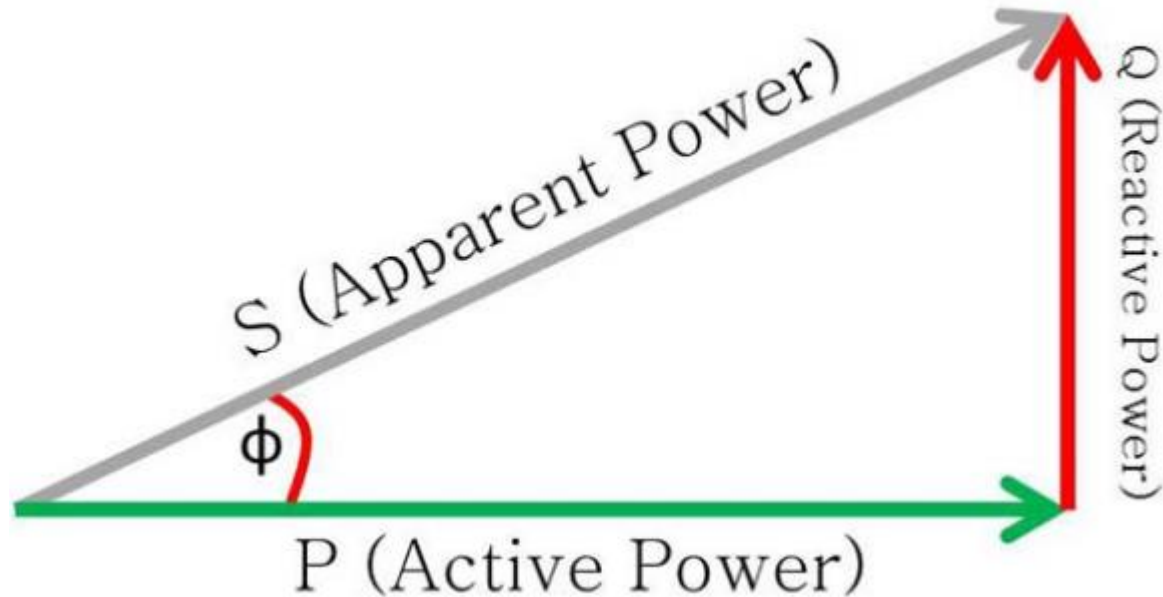
- For a three phase electrical circuit, Apparent power (kVA) = $\sqrt{3}$ x Voltage x Amperes / 1000
- Reactive Power (kVAr)
 - Reactive power is the portion of apparent power that does no work.
 - This type of power must be supplied to all types of magnetic equipment, such as motors, transformers etc.
 - Larger the magnetizing requirement, larger the kVAr.
- Active Power, Kilowatt (kW)
 - kW is the active power or the work-producing part of apparent power.
 - For single phase, Power (kW) = Voltage x Amperes x Power factor / 1000
 - For three phase, Power (kW) = $\sqrt{3}$ x Voltage x Amperes x Power factor / 1000
- Power Factor
 - Power Factor (PF) is the ratio between the active power (kW) and apparent power (kVA).

Electrical Energy Basics



– Power Factor, $\cos \phi = \frac{\text{Active power (kW)}}{\text{Apparent power (kVA)}} = \frac{\text{kW}}{\sqrt{(\text{kW})^2 + (\text{kVAr})^2}}$
 $= 1.0$ (when $\text{kVAr} = 0$).

POWER TRIANGLE



Electrical Energy Basics

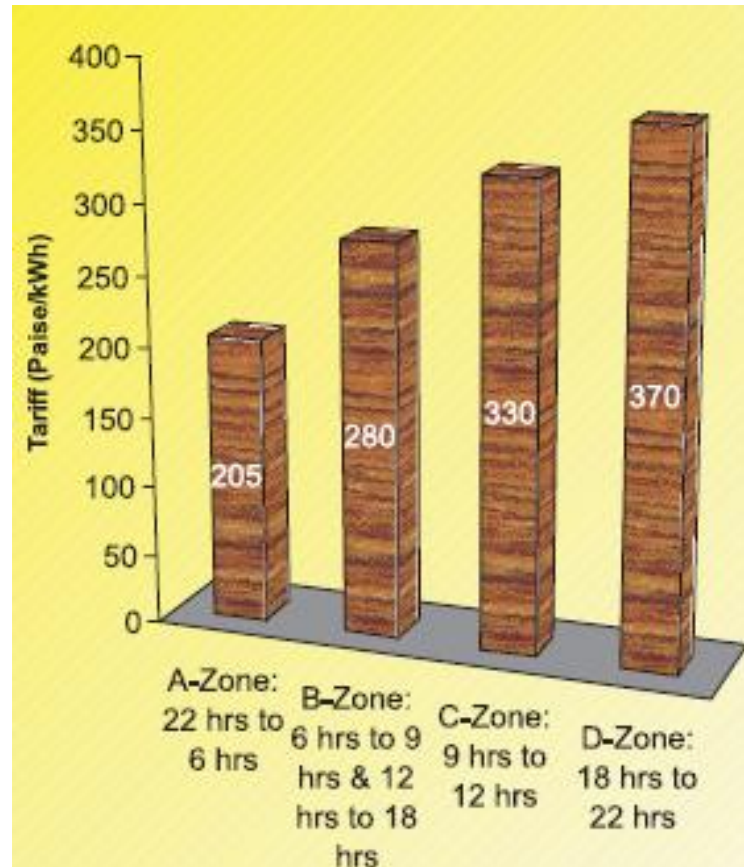


- Electricity unit, kilowatt-hour (kWh)
 - Kilowatt-hour is the energy consumed by 1000 Watts in one hour.
 - If 1kW (1000 watts) of a electrical equipment is operated for 1 hour, it would consume 1 kWh of energy (1 unit of electricity).
 - For a company, it is the amount of electrical units in kWh recorded in the plant over a month for billing purpose.
 - The company is charged / billed based on kWh consumption.
- Electricity Tariff
 - Electrical utility or power supplying companies charge industrial customers not only based on the amount of energy used (kWh) but also on the peak demand (kVA) for each month.
- Time of Day (TOD) Tariff
 - Many electrical utilities like to have flat demand curve to achieve high plant efficiency.

Electrical Energy Basics



- Users are encouraged to draw more power during off-peak hours (say during night time) and less power during peak hours.
- As per their plan, they offer TOD Tariff, which may be incentives or disincentives.
- Energy meter will record peak and non-peak consumption separately by timer control.
- TOD tariff gives opportunity for the user to reduce their billing, as off peak hour tariff charged are quite low in comparison to peak hour tariff.



Electrical Energy Basics – Problem (1)



- A 3-phase AC induction motor (20 kW capacity) is used for pumping operation. Electrical parameter such as current, volt and power factor were measured with power analyzer. Find energy consumption of motor in one hour? (line volts. = 440 V, line current = 25 amps and PF = 0.90).
- Given: Voltage = 440 V, Current = 25 A, PF = 0.9
- For three phase, Power (kW) = $\sqrt{3} \times \text{Voltage} \times \text{Amperes} \times \text{Power factor} / 1000$
- Power consumption = 17.147 kW
- Energy consumption in one hour
= 17.147 kW x 1 hour = **17.147 kWh**

Thermal Energy Basics



- Temperature and pressure are measures of the physical state of a substance which are closely related to the energy contained in the substance.
- Temperature
 - It is the degree of hotness or coldness measured on a definite scale.
 - Heat is a form of energy.
 - Temperature is a measure of its thermal effects.
 - In other words, temperature is a means of determining sensible heat content of the substance.
- Pressure
 - It is the force per unit area applied to outside of a body.
 - When we heat a gas in a confined space, we create more force; a pressure increase.
 - For example, heating the air inside a balloon will cause the balloon to stretch as the pressure increases.
 - Pressure is indicative of stored energy.
 - Steam at high pressures contains much more energy than at low pressures.

Temperature Scales



- **Fahrenheit and Celsius scales**

- Based on a specification of the number of increments between the freezing point and boiling point of water at standard atmospheric pressure.
 - The Celsius scale has 100 units between these points, while the Fahrenheit scale has 180 units.

- **Kelvin and Rankine scales**

- The absolute Celsius scale is called the Kelvin scale
- The absolute Fahrenheit scale is termed the Rankine scale.
- The zero points on both absolute scales represent the same physical state
- The ratio of two values is the same, regardless of the absolute scale used.

K	°C	°F	°R
2273.16	2000	3632	4091.69
1773.16	1500	2732	3191.69
1273.16	1000	1832	2291.69
773.16	500	932	1391.69
673.16	400	752	1211.69
573.16	300	572	1031.69
473.16	200	392	851.69
373.16	100	212.0	671.69
273.16	0	32.0	491.69
233.16	-40	-40	419.69
173.16	-100	-148	311.69

$$^{\circ}\text{F} = 32.0 + \frac{9}{5}^{\circ}\text{C}$$

$$^{\circ}\text{R} = \frac{9}{5}\text{K}$$

$$\left(\frac{T_2}{T_1}\right)_{\text{Rankine}} = \left(\frac{T_2}{T_1}\right)_{\text{Kelvin}}$$

Thermal Energy Basics



- Heat
 - Heat is a form of energy, a distinct and measurable property of all matter.
 - The quantity of heat depends on the quantity and type of substance involved.
- Unit of Heat
 - Calorie is one of the units for measuring the quantity of heat. It is the quantity of heat, which can raise the temperature of 1 g of water by 1°C .
 - SI unit is joule (J)
- Specific Heat
 - If the same amount of heat energy is supplied to equal quantities of water and milk, their temperature goes up by different amounts. This is because of the property called the specific heat of a substance
 - It is defined as the quantity of heat required to raise the temperature of 1kg of a substance through 1°C .

Thermal Energy Basics



- The specific heat of water is very high as compared to other common substances; it takes a lot of heat to raise the temperature of water.
- Also, when water is cooled, it gives out a large quantity of heat.

TABLE 2.1 SPECIFIC HEAT OF SOME COMMON SUBSTANCES	
Substance	Specific Heat (Joules / kg °C)
Lead	130
Mercury	140
Brass	380
Copper	390
Iron	470
Glass	670
Aluminium	910
Rubber	1890
Ice	2100
Alcohol	2400
Water	4200

Thermal Energy Basics



- Sensible heat
 - It is that heat which when added or subtracted to a substance results in a change of its temperature without any phase change.
- Quantity of Heat
 - The quantity of heat, Q , required to increase the temperature of a substance depends on
 - mass of the substance (m)
 - increase in temperature (ΔT)
 - specific heat of the substance (C_p)
 - The quantity of heat is given by:
 $Q = \text{mass} \times \text{specific heat} \times \text{increase in temperature}$
 $Q = m \times C_p \times \Delta T$
- Phase Change
 - Fusion – The change of state from solid state to liquid state at melting point temperature

Thermal Energy Basics



- Vaporization – The change of state from liquid state to gas state at boiling point temperature.
- Latent heat
 - It is the change in heat content of a substance, when its physical state is changed without a change in temperature.
- Latent heat of fusion
 - The latent heat of fusion of a substance is the quantity of heat required to convert 1kg solid to liquid state without change of temperature.
 - It is represented by the symbol L .
 - $L(\text{ice}) = 336 \text{ kJ/kg}$
- Latent Heat of Vaporization
 - The latent heat of vaporization of a substance is the quantity of heat required to change 1kg of the substance from liquid to vapour state without change of temperature.
 - It is also denoted by the symbol L and its unit is also J/kg .

Thermal Energy Basics



- The latent heat of vaporization of water is 22,60,000 J/kg.
- When 1 kg of steam at 100°C condenses to form water at 100°C, it gives out 2260 kJ (540 kCals) of heat.
- Steam gives out more heat than an equal amount of boiling water because of its latent heat.
- Latent heat
 - It is the change in heat content of a substance, when its physical state is changed without a change in temperature.
- Super Heat
 - The heating of vapour, particularly saturated steam to a temperature much higher than the boiling point at the existing pressure.
 - This is done in power plants to improve efficiency and to avoid condensation in the turbine.
- Humidity
 - Refers to the moisture content of air

Thermal Energy Basics



- Humidity or Absolute humidity
 - The mass of water vapour per unit mass of dry air (vapour-free gas).
- Specific humidity
 - The mass of water vapour per unit mass of vapour-gas mixture.
- Relative Humidity (RH)
 - It is the measure of degree of saturation of the air at a given temperature.
 - Relative humidity given as a percentage is the actual water content of the air divided by the moisture content of fully saturated air at the existing temperature.
- Dew Point Temperature
 - It is the temperature at which condensation of water vapour from the air just begins as the temperature of the air-water vapour mixture falls.
 - This is the temperature at which a vapor-gas mixture becomes saturated when cooled at constant total pressure out of contact with a liquid.
 - It is a measure of the latent heat content of air-water-vapour mixtures.

Thermal Energy Basics



- Dry bulb Temperature
 - It is an indication of the sensible heat content of air-water vapour mixtures.
 - The true temperature of the air-water vapour mixture.
 - Temperature measured by a thermometer whose bulb is dry.
- Wet bulb Temperature
 - It is a measure of total heat content or enthalpy.
 - The wet-bulb temperature is the steady-state temperature reached by a small amount of liquid evaporating into a large amount of unsaturated vapor-gas mixture.
 - A thermometer whose bulb is covered with a wick kept wet with the liquid is immersed in a rapidly moving stream of the gas mixture to measure the wet-bulb temperature.
 - It can be used to measure the humidity of the mixture.

Thermal Energy Basics



- Fuel Density
 - Density is the ratio of the mass of the fuel to the volume of the fuel at a stated temperature.
- Specific gravity of fuel
 - The density of fuel, relative to water, is called specific gravity.
 - The specific gravity of water is defined as 1.
 - As it is a ratio there are no units.
 - Higher the specific gravity, higher will be the heating values.
- Viscosity
 - The viscosity of a fluid is a measure of its internal resistance to flow.
 - All liquid fuels decrease in viscosity with increasing temperature
- Calorific Value
 - Energy content in an organic matter (Calorific Value) can be measured by burning it and measuring the heat released.

Thermal Energy Basics



- **Calorific Value**

- Energy content in an organic matter (Calorific Value) can be measured by burning it and measuring the heat released.
- This is done by placing a sample of known mass in a bomb calorimeter, a device that is completely sealed and insulated to prevent heat loss.
- A thermometer is placed inside (but it can be read from the outside) and the increase in temperature after the sample is burnt completely is measured.
- From this data, energy content in the organic matter can be found out.
- The heating value of fuel is the measure of the heat released during the complete combustion of unit weight of fuel.
- It is expressed as Gross Calorific Value (GCV) or Net Calorific Value (NCV).
- The difference between GCV and NCV is the heat of vaporization of the moisture and atomic hydrogen (conversion to water vapour) in the fuel.
- Typical GCV and NCV for heavy fuel oil are 10,500 kcal/kg and 9,800 kcal/kg.

Thermal Energy Basics



- Heat Transfer
 - Heat will always be transferred from higher temperature to lower temperature independent of the mode.
 - The energy transferred is measured in Joules (kcal or Btu).
 - The rate of energy transfer, more commonly called heat transfer, is measured in Joules/second (kcal/hr or Btu/hr).
 - Heat is transferred by three primary modes:
 - Conduction (Energy transfer in a solid)
 - Convection (Energy transfer in a fluid)
 - Radiation (Does not need a material to travel through)
- Conduction
 - The conduction of heat takes place, when two bodies are in contact with one another.

Thermal Energy Basics



- Conduction

- The conduction of heat takes place, when two bodies are in contact with one another.
- If one body is at a higher temperature than the other, the motion of the molecules in the hotter body will vibrate the molecules at the point of contact in the cooler body and consequently result in increase in temperature.
- The amount of heat transferred by conduction depends upon
 - the temperature difference
 - the properties of the material involved
 - the thickness of the material
 - the surface contact area
 - the duration of the transfer
- Good conductors of heat are typically substances that are dense as they have molecules close together.
- This allows the molecular agitation process to permeate the substance easily.

Thermal Energy Basics



- Metals are good conductors of heat, while gaseous substance, having low densities or widely spaced molecules, are poor conductors of heat.
- Poor conductors of heat are usually called insulators.
- The measure of the ability of a substance to insulate is its thermal resistance, the inverse of the thermal conductivity, the ability to conduct heat.
- Typical units of measure for conductive heat transfer are:
 - Per unit area (for a given thickness) : Watt per square meter (W/m^2)
 - Overall : Watt (W) or kilowatts (kW)
- Convection
 - The transfer of heat by convection involves the movement of a fluid such as a gas or liquid from the hot to the cold portion.
 - Typical units of measure for convective heat transfer are Watt (W) or kilowatts (kW)
 - There are two types of convection:
 - Natural convection
 - Forced convection

Thermal Energy Basics



- Natural convection
 - In case of natural convection, the fluid in contact with or adjacent to a high temperature body is heated by conduction.
 - As it is heated, it expands, becomes less dense and consequently rises.
 - This begins a fluid motion process in which a circulating current of fluid moves past the heated body, continuously transferring heat away from it.
- Forced convection
 - In the case of forced convection, the movement of the fluid is forced by a fan, pump or other external means.
 - A centralized hot air heating system is a good example of forced convection.
 - Convection depends on the thermal properties of the fluid as well as surface conditions at the body and other factors that affect the ability of the fluid to flow.
 - With a low conductivity fluid such as air, a rough surface can trap air against the surface reducing the conductive heat transfer and consequently reducing the convective currents.

Thermal Energy Basics



- Thermal Radiation

- Thermal radiation is a process in which energy is transferred by electromagnetic waves similar to light waves.
- These waves may be both visible (light) and invisible.
- A very common example of thermal radiation is a heating element on a heater.
- When the heater element is first switched on, the radiation is invisible, but you can feel the warmth it radiates.
- As the element heats, it will glow orange and some of the radiation is now visible.
- The hotter the element, the brighter it glows and the more radiant energy it emits.
- The key processes in the interaction of a substance with thermal radiation are Absorption, transmission, and reflection.

Thermal Energy Basics



- The key processes in the interaction of a substance with thermal radiation are:
 - Absorption
 - The process by which radiation enters a body and becomes heat
 - Transmission
 - The process by which radiation passes through a body
 - Reflection
 - The process by which radiation is neither absorbed or transmitted through the body; rather it bounces off.
- Objects receive thermal radiation when they are struck by electromagnetic waves, thereby agitating the molecules and atoms.
- More agitation means more energy and a higher temperature.
- Energy is transferred to one body from another without contact or transporting medium such as air or water.
- In fact, thermal radiation heat transfer is the only form of heat transfer possible in a vacuum.

Thermal Energy Basics



- All bodies emit a certain amount of radiation.
- The amount depends upon the body's temperature and nature of its surface.
- Some bodies only emit a small amount of radiant energy for their temperature, commonly called low emissivity materials (abbreviated low-E).
- Low-E windows are used to control the heat radiation in and out of buildings.
- Windows can be designed to reflect, absorb and transmit different parts of the sun's radiant energy.
- The condition of a body's surface will determine the amount of thermal radiation that is absorbed, reflected or re-emitted.
- Surfaces that are black and rough, such as black iron, will absorb and re-emit almost all the energy that strikes them.
- Polished and smooth surfaces will not absorb, but reflect, a large part of the incoming radiant energy.

Thermal Energy Basics



- **Evaporation**
 - The change by which any substance is converted from a liquid state and carried off as vapour.
 - Example: People are cooled by evaporation of perspiration from the skin and refrigeration is accomplished by evaporating the liquid refrigerant.
 - Evaporation is a cooling process.
- **Condensation**
 - The change by which any substance is converted from a gaseous state to liquid state.
 - Example: Condensation on the other hand is a heating process.
 - As molecules of vapour condense and become liquid, their latent heat of vapourisation evidences itself again as sensible heat, indicated by a rise in temperature.
 - This heating effect of condensation is what causes a rise in atmospheric temperature often noted as fog forms and as rain or snow begins to fall.

Thermal Energy Basics



- Steam
 - Steam has been a popular mode of conveying energy, since the industrial revolution.
 - The following characteristics of steam make it so popular and useful to the industry:
 - High specific heat and latent heat
 - High heat transfer coefficient
 - Easy to control and distribute
 - Cheap and inert
 - Steam is used for generating power and also used in process industries, such as, sugar, paper, fertilizer, refineries, petrochemicals, chemical, food, synthetic fibre and textiles.

Thermal Energy Basics



- Cogeneration
 - In the process industries, the high pressure steam produced in the boiler, is first expanded in a steam turbine for generating power.
 - The extraction or bleed from the turbine, which are generally at low pressure, are used for the process.
 - This method of producing power, by using the steam generated for process in the boiler, is called "Cogeneration."

Thermal Energy Basics



- Cogeneration
 - In the process industries, the high pressure steam produced in the boiler, is first expanded in a steam turbine for generating power.
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Problem (2)



- 10 kg of steam at 100°C with latent heat of vapourisation 2260 kJ/kg is cooled to 50°C. If the specific heat of water is 4200 J/kg°C, find the quantity of heat given out.
- Answer: $2260 \times 10 + 4.2 \times 10 \times 50 = \mathbf{24700 \text{ kJ}}$

Problem (3)



- A fluorescent tube light consumes 40 W for the tube and 10 W for choke. If the lamp operates for 8 hours a day for 300 days in a year, calculate the total energy cost per annum if the energy cost is Rs.3/- per kWh
- Answer: $[(40+10) \times 8 \times 300 / 1000] \times 3 = \text{Rs. } 360$